

B.Tech, 4th SEMESTER
ESE EXAMINATION AI (BLENDED), EVEN SEM. - 2025-26
AI Driven Full Stack Development (AI308B)

Time: 3 Hours

Total Marks: 50

Note: The project report in PDF format must be uploaded on Moodle during the examination. Part D

CASE STUDY: AI-Based Employee Performance Analytics & Recommendation System

Design and develop a full-stack MERN application that analyzes employee performance data and provides AI-powered recommendations using an external AI API (OpenRouter/OpenAI compatible API).

The system should allow HR/Admin users to:

- Add and manage employee details
- Track employee skills and performance metrics
- Generate AI-based recommendations for promotions/training
- View employee analytics and rankings
- Secure the application using authentication

Q1	All parts are compulsory	CO-3	
	FRONTEND REQUIREMENTS	BL	Marks
	Create the following components: 1. Employee Registration Form 2. Employee List Page 3. Search & Filter Section 4. AI Recommendation Display Page Evaluation Criteria • Proper React Component Structure • useState & useEffect usage • Form Handling • Responsive UI • API Integration using Axios/Fetch Store employee details: • Employee Name • Email • Department • Skills (Array) • Performance Score • Years of Experience	3	8

Q2	All parts are compulsory	CO-2	
	BACKEND REQUIREMENTS (Node.js & Express.js)	BL	Marks
	Implement: 1. REST APIs 2. Controller & Route Structure 3. Validation Logic 4. Error Handling Middleware Required API Endpoints Employee APIs a. Add Employee POST /api/employees Sample Body <pre>{ "name": "Aman Verma", "email": "aman@gmail.com", "department": "Development", "skills": ["React", "Node.js", "MongoDB"], "performanceScore": 85, "experience": 3 }</pre> b. Get All Employees GET /api/employees c. Search Employee GET /api/employees/search?department=Development AI APIs d. AI Recommendation API	3	8

	POST /api/ai/recommend												
Q3	All parts are compulsory	CO-2											
	DATABASE IMPLEMENTATION (MongoDB)	BL	Marks										
	Perform: 1. Schema Creation 2. CRUD Operations 3. Query Filtering 4. Data Validation Test Cases<table><tr><th>Test Case</th><th>Expected Output</th></tr><tr><td>Insert valid employee</td><td>Employee stored successfully</td></tr><tr><td>Duplicate email</td><td>Error message</td></tr><tr><td>Missing performance score</td><td>Validation error</td></tr><tr><td>Search by department</td><td>Filtered employee list</td></tr></table>	Test Case	Expected Output	Insert valid employee	Employee stored successfully	Duplicate email	Error message	Missing performance score	Validation error	Search by department	Filtered employee list	3	6
Test Case	Expected Output												
Insert valid employee	Employee stored successfully												
Duplicate email	Error message												
Missing performance score	Validation error												
Search by department	Filtered employee list												
Q4	All parts are compulsory	CO-4											
	MERN INTEGRATION (Frontend–Backend–Database)	BL	Marks										
	Integrate: • React Frontend with Express APIs • MongoDB Data Fetching • Dynamic Data Rendering Test Cases<table><tr><th>Action</th><th>Expected Result</th></tr><tr><td>Add employee from frontend</td><td>Data saved in MongoDB</td></tr><tr><td>Fetch employee list</td><td>Display all employees</td></tr><tr><td>Update performance score</td><td>Updated data shown</td></tr><tr><td>Delete employee</td><td>Employee removed successfully</td></tr></table>	Action	Expected Result	Add employee from frontend	Data saved in MongoDB	Fetch employee list	Display all employees	Update performance score	Updated data shown	Delete employee	Employee removed successfully	3	6
Action	Expected Result												
Add employee from frontend	Data saved in MongoDB												
Fetch employee list	Display all employees												
Update performance score	Updated data shown												
Delete employee	Employee removed successfully												

Q5	All parts are compulsory	CO-5											
	AI INTEGRATION	BL	Marks										
	Integrate OpenRouter/OpenAI compatible API. AI Features 1. Promotion Recommendation 2. Employee Ranking 3. Training Suggestions 4. AI Feedback Generation Test Cases <table><tr><th>Input</th><th>Expected AI Output</th></tr><tr><td>High performance employee</td><td>Promotion suggestion</td></tr><tr><td>Low score employee</td><td>Improvement feedback</td></tr><tr><td>Employee with missing skills</td><td>Skill enhancement recommendation</td></tr><tr><td>Multiple employees</td><td>Ranked recommendations</td></tr></table>	Input	Expected AI Output	High performance employee	Promotion suggestion	Low score employee	Improvement feedback	Employee with missing skills	Skill enhancement recommendation	Multiple employees	Ranked recommendations	3	8
Input	Expected AI Output												
High performance employee	Promotion suggestion												
Low score employee	Improvement feedback												
Employee with missing skills	Skill enhancement recommendation												
Multiple employees	Ranked recommendations												
Q6	All parts are compulsory	CO-3											
	AUTHENTICATION & SECURITY	BL	Marks										
	Implement: 1. JWT Authentication 2. Password Hashing using bcrypt 3. Protected Routes 4. Login & Signup APIs Test Cases <table><tr><th>Test Case</th><th>Expected Output</th></tr><tr><td>Valid Login</td><td>JWT Token generated</td></tr><tr><td>Invalid Password</td><td>Unauthorized error</td></tr><tr><td>Access protected route without token</td><td>Access denied</td></tr><tr><td>Password stored in DB</td><td>Encrypted format</td></tr></table>	Test Case	Expected Output	Valid Login	JWT Token generated	Invalid Password	Unauthorized error	Access protected route without token	Access denied	Password stored in DB	Encrypted format	3	5
Test Case	Expected Output												
Valid Login	JWT Token generated												
Invalid Password	Unauthorized error												
Access protected route without token	Access denied												
Password stored in DB	Encrypted format												

Q7	All parts are compulsory	CO-1	
	GIT & GITHUB USAGE	BL	Marks
	Evaluation - Clean commit history - Project documentation - Proper repository structure	3	3
Q8	All parts are compulsory	CO-4	
	DEPLOYMENT ON RENDER	BL	Marks
	Deploy: - Frontend - Backend - MongoDB Connection Submission Requirements - Live Frontend URL - Backend API URL - GitHub Repository Link	3	3
Q9	All parts are compulsory	CO-1	
	CODE QUALITY & DOCUMENTATION	BL	Marks
	Evaluation Criteria - Folder Structure - Naming Conventions - Comments & Documentation - Reusable Components	3	3

NOTE: Make Docs and convert it into PDF for submission and it should contain:

- o Code
- o Code output (screenshots)
- o Postman/Thunder client all HTTP request screenshots
- o MongoDB items storage part (screenshots)
- o Render deployment successful (screenshot)
- o Render testing live URL for each end-point